

any AWS CCI partner for setting up the contact centre, and integrate it with other AWS native services like data analytics, ML based data processing and data storage, to name a few.

Azure Cognitive Services

Microsoft's Azure Cognitive Services can help you realise partial or full automation of telephony based customer interactions, and provide accessibility across multiple channels. With its language and speech services, you can analyse call centre transcriptions, extract and redact personally identifiable information (PII), summarise the transcription, and detect the sentiments.

Azure Cognitive Services can be implemented in call and contact centres in various ways.

- **Virtual agents:** Conversational AI based telephony-integrated voice bots and voice-enabled chatbots.
- **Agent assistance:** Real-time transcription and analysis of a call to improve the customer experience by providing insights and suggesting actions to agents.
- **Post-call analytics:** Post-call analysis for creating insights into customer conversations to improve understanding and support continuous improvement of call handling.

Google Contact Center AI

Google Cloud Platform (GCP) provides loads of AI based solutions, which are integrated with Google Contact Center AI services for virtual assistance. GCP's Contact Center AI is a hybrid solution suitable for on-premises private as well as the public cloud environment.

Some of its important components, which are now in general availability (GA), are listed below.

Dialogflow CX: This helps with handling complex customer queries and supplemental questions by looking for the required responses from a knowledge base. It supports human agents by providing the relevant details required to respond to a query. Dialogflow CX agents build on Google's natural language understanding (NLU) algorithms to handle these search capabilities.

Agent Assistant: This is built with AI assisted conversational services to prepare the recommended answer choices for human agents to choose from, for replying to customer queries. It uses a centralised knowledge base to prepare ready-to-use responses. It transcribes live calls in different languages and helps train AI models to analyse frequently asked questions, answer patterns and customer groups based on interest.

CCAI Insights: These services include text-to-speech and speech-to-text conversion, sentiment analysis, ranking of responses, integrated IVR services, and natural language processing capabilities to provide unified customer support powered by Google Cloud AI services.

Contact centre AI: The trends

Contact centres are undergoing a massive transformation. During the Covid-19 pandemic, call volumes increased but the number of customer service agents on duty fell. Calls increased by nearly 300 per cent during the first few months of the pandemic, resulting in longer hold times and more escalations to human agents or other support representatives.

With about 90 per cent of contact centres migrating to remote and

hybrid workplace models since the beginning of the pandemic, companies are investing in new technologies that allow them to provide high quality and personalised customer service experiences.

In response to the changing landscape, many companies are turning to AI to power customer service operations. Listed below are six trends with respect to the use of AI in contact centres that promise to transform customer service and experience.

Alleviation of shortages in staff: Labour shortages were a side effect of the pandemic, and many industries found it difficult to manage high call volumes with low volumes of agents. Many businesses are turning to AI to account for the gaps within their workforces.

AI is helping to fill skills shortages of the existing workforce through career transition support tools. It is also helping employees do their jobs better and faster using digital assistants and in-house AI-driven training programmes.

Understanding the nuances of human conversations: AI solutions that are perceptive to caller intent, language, and sentiment are vital. Contact centre AI helps to understand the nuances of human conversation because it can pick up on the tone of voice, inflection, etc, to detect mood and modify behaviour accordingly. Understanding voice nuances and serving customers on that basis improves the overall experience.

Self-service: Self-service helps to deliver customer support quickly and efficiently since customers don't have to wait for a human agent. Companies continue to provide human backup, but prioritise AI support for their contact centres. One way businesses can use AI to deliver round-the-clock support is through chatbots and interactive voice response (IVR) systems.

Chatbots automate a personalised experience, connect meaningfully with

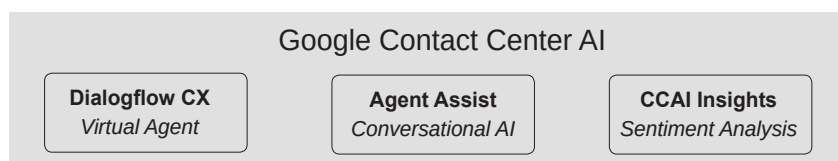


Figure 1: Google Contact Center AI